

Ex 2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

```
from google.colab import files
import pandas as pd
uploaded = files.upload()
df = pd.read_csv("sales_data.csv")
print(df)
```

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Saving sales_data.csv to sales_data (2).csv

	s.no	day	product	name	sales	quantity	region
0	1	Monday	Laptop	Arun	50000	2	North
1	2	Tuesday	Mobile	Bala	30000	3	South
2	3	Wednesday	Keyboard	Cathy	2500	10	East
3	4	Thursday	Mouse	Dinesh	1200	15	West
4	5	Friday	Monitor	Elena	18000	4	North
5	6	Saturday	Tablet	Farhan	22000	5	South
6	7	Sunday	Printer	Gokul	15000	3	East
7	8	Monday	Headphones	Hari	5000	8	West
8	9	Tuesday	Laptop	Isha	55000	2	North
9	10	Wednesday	Mobile	Jeeva	32000	3	South

	s.no	day	product	name	sales	quantity	region
0	1	Monday	Laptop	Arun	50000	2	North
1	2	Tuesday	Mobile	Bala	30000	3	South
2	3	Wednesday	Keyboard	Cathy	2500	10	East
3	4	Thursday	Mouse	Dinesh	1200	15	West
4	5	Friday	Monitor	Elena	18000	4	North

```
from google.colab import files
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
uploaded = files.upload()
df = pd.read_csv("sale_data.csv")
print(df)
df['sales'] = df['sales'].fillna(df['sales'].mean())
df['quantity'] = df['quantity'].fillna(df['quantity'].mean())
print(df)
df = df.dropna()
print(df)
```

```
print(df.describe(include='all'))
result = df.groupby('product')[['sales', 'quantity']].sum()
print(result)
total_sales = df.groupby('product')['sales'].sum()
print(total_sales)
total_sales.plot(kind='bar')
plt.xlabel('product')
plt.ylabel('total sales')
plt.title('total sales by product')
plt.show()
df.groupby('day')['sales'].sum().plot(kind='line', marker='o')
plt.xlabel('Day')
plt.ylabel('Total Sales')
plt.show()
pivot = pd.pivot_table(
    df,
    index='day',
    columns='product',
    values='sales',
    aggfunc='sum'
)

print(pivot)
corr = df.corr(numeric_only = True)
sns.heatmap(corr,annot=True,cmap='coolwarm')
plt.title("correlation matrix")
plt.show()
```

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Saving sale_data.csv to sale_data (11).csv

	s.no	day	product	name	sales	quantity	region
0	1	Monday	Laptop	Arun	50000.0	2.0	North
1	2	Tuesday	Mobile	Bala	NaN	3.0	South
2	3	Wednesday	NaN	Cathy	2500.0	NaN	East
3	4	NaN	Mouse	Dinesh	1200.0	15.0	West
4	5	Friday	Monitor	NaN	18000.0	4.0	NaN
5	6	Saturday	Tablet	Farhan	NaN	5.0	South
6	7	Sunday	Printer	Gokul	15000.0	NaN	East
7	8	Monday	NaN	Hari	5000.0	8.0	West
8	9	Tuesday	Laptop	Isha	55000.0	2.0	North
9	10	Wednesday	Mobile	NaN	32000.0	3.0	NaN

	s.no	day	product	name	sales	quantity	region
0	1	Monday	Laptop	Arun	50000.0	2.00	North
1	2	Tuesday	Mobile	Bala	22337.5	3.00	South
2	3	Wednesday	NaN	Cathy	2500.0	5.25	East
3	4	NaN	Mouse	Dinesh	1200.0	15.00	West
4	5	Friday	Monitor	NaN	18000.0	4.00	NaN
5	6	Saturday	Tablet	Farhan	22337.5	5.00	South
6	7	Sunday	Printer	Gokul	15000.0	5.25	East
7	8	Monday	NaN	Hari	5000.0	8.00	West
8	9	Tuesday	Laptop	Isha	55000.0	2.00	North
9	10	Wednesday	Mobile	NaN	32000.0	3.00	NaN

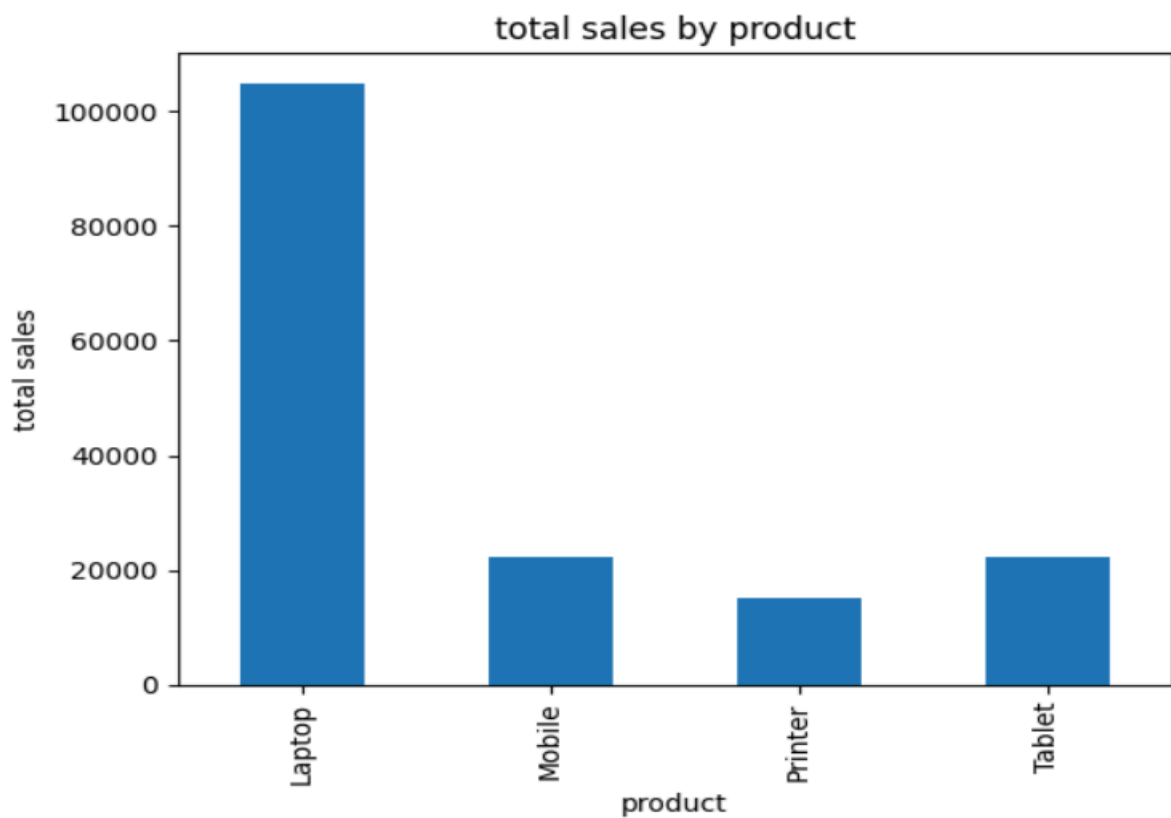
	s.no	day	product	name	sales	quantity	region
0	1	Monday	Laptop	Arun	50000.0	2.00	North
1	2	Tuesday	Mobile	Bala	22337.5	3.00	South
5	6	Saturday	Tablet	Farhan	22337.5	5.00	South
6	7	Sunday	Printer	Gokul	15000.0	5.25	East
8	9	Tuesday	Laptop	Isha	55000.0	2.00	North

```

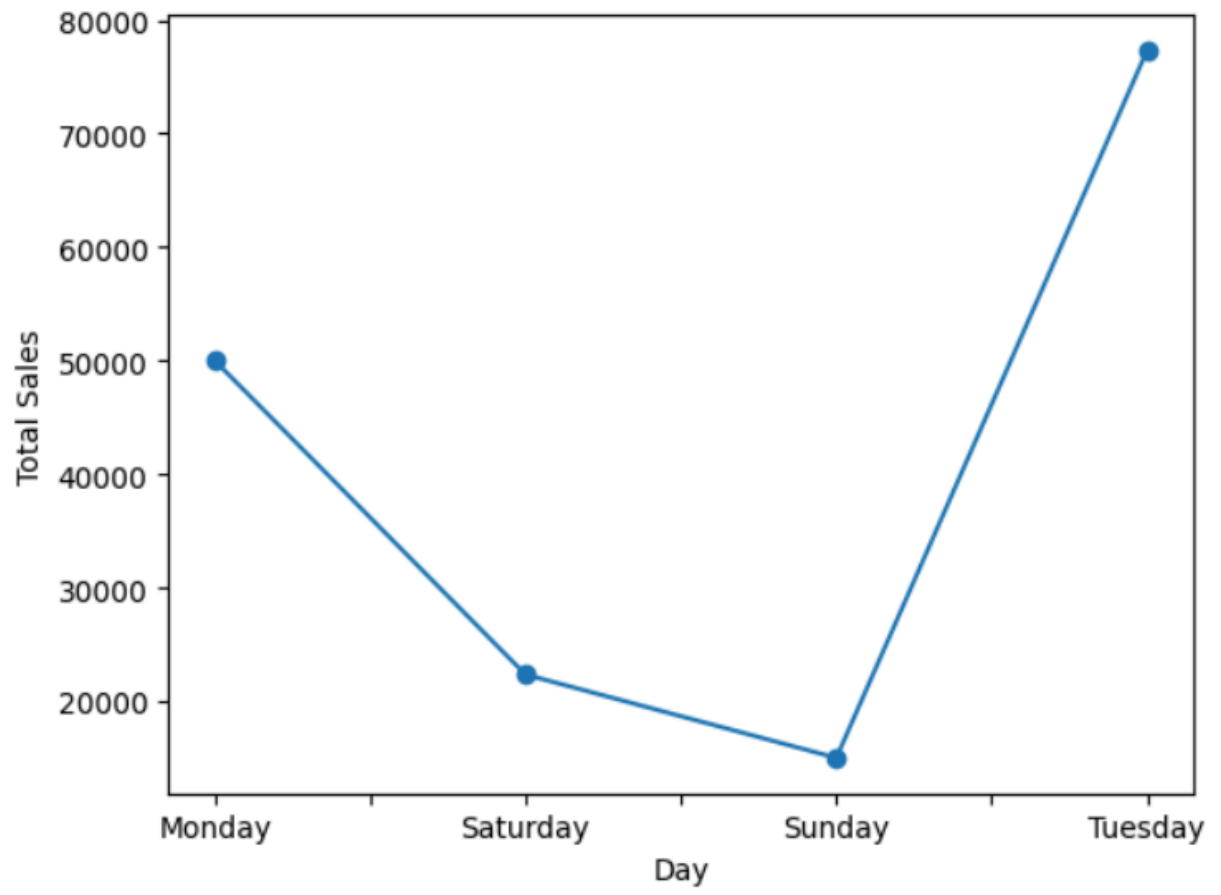
... count      s.no      day product name      sales      quantity      region
unique      NaN      4      4      5      NaN      NaN      3
top      NaN      Tuesday      Laptop      Arun      NaN      NaN      North
freq      NaN      2      2      1      NaN      NaN      2
mean      5.000000      NaN      NaN      NaN      32935.000000      3.450000      NaN
std      3.391165      NaN      NaN      NaN      18195.855898      1.585087      NaN
min      1.000000      NaN      NaN      NaN      15000.000000      2.000000      NaN
25%      2.000000      NaN      NaN      NaN      22337.500000      2.000000      NaN
50%      6.000000      NaN      NaN      NaN      22337.500000      3.000000      NaN
75%      7.000000      NaN      NaN      NaN      50000.000000      5.000000      NaN
max      9.000000      NaN      NaN      NaN      55000.000000      5.250000      NaN

      sales      quantity
product
Laptop      105000.0      4.00
Mobile      22337.5      3.00
Printer      15000.0      5.25
Tablet      22337.5      5.00
product
Laptop      105000.0
Mobile      22337.5
Printer      15000.0
Tablet      22337.5
Name: sales, dtype: float64

```



product



```
product    Laptop    Mobile    Printer    Tablet
day
... Monday    50000.0      NaN      NaN      NaN
Saturday     NaN      NaN      NaN    22337.5
Sunday       NaN      NaN    15000.0      NaN
Tuesday     55000.0    22337.5      NaN      NaN
```

